

REGIME-DEPENDENT VOLATILITY SENSITIVITY

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ABSTRACT

This study examines whether regime-dependent volatility sensitivity and its impact on trend-following profitability are universal across asset classes. Using a common sample of daily data for Bitcoin, Gold, and the SCP 500 from 2015 to 2026, we evaluate a standardized volatility-targeted EMA trend strategy under regime-conditional stress tests. To account for liquidity collapses, we implement dynamic transaction cost modeling, penalizing execution friction at 3x the baseline during high-volatility expansions. We document a structural divergence: net of stressed transaction costs, Bitcoin maintains a statistically invariant and robust performance profile across volatility regimes, whereas traditional assets suffer severe degradation. Statistical tests confirm that trend performance is highly regime-sensitive for traditional assets ($p < 0.001$). For Bitcoin, we fail to reject the null hypothesis ($p = 0.195$), and supporting Bayes Factor analysis confirms the equivalence of trend performance across volatility states. These findings provide empirical evidence consistent with the hypothesis that volatility is an asset-dependent signal rather than a homogeneous risk factor. Consequently, portfolio construction frameworks that mechanically de-leverage during volatility expansions—such as volatility targeting or risk parity—may be structurally misaligned for assets exhibiting positive volatility-return dependence. We propose conditioning algorithmic exposure on asset-specific volatility characteristics rather than applying universal defensive filters.

KEYWORDS: Time-Series Momentum; Regime-Dependent Volatility Sensitivity; Volatility Targeting; Cryptocurrency Microstructure; Risk Parity; Transaction Cost Analysis; Asset Heterogeneity; Trend Following.

JEL CLASSIFICATION : C58 , C11 , G11 , G17

AMS CLASSIFICATION: 91G10, 62P05, 91G70

INTRODUCTION

Time-series momentum remains one of the most robust documented anomalies in financial markets. Its appeal to institutional investors lies in its convex payoff profile and its historical ability to provide "crisis alpha"—generating positive returns during extended periods of equity market distress (Harvey et al., 2018). Standard implementations, however, implicitly assume that volatility is uniformly harmful to signal quality and therefore requires defensive suppression through algorithmic filtering and position de-leveraging.

This assumption is increasingly questioned in the context of digital asset markets. Bitcoin, in particular, exhibits extreme kurtosis, non-normal return distributions, and persistent structural breaks. Furthermore, recent literature documents shifting correlation dynamics between cryptocurrencies and traditional safe-haven assets post-2020 (e.g., Liu, Tsyvinski, C Wu, 2022), which severely complicates the application of universal risk frameworks. Practitioners frequently report that standard trend-following techniques underperform during high-volatility periods in crypto markets, which has motivated a flight to complexity using increasingly sophisticated signal-processing techniques.

Yet mathematical complexity does not necessarily resolve structural mismatch. This paper evaluates a simpler, more fundamental hypothesis: the relationship between volatility and trend-following

profitability is inherently asset-dependent. We introduce the concept of **Regime-Dependent Volatility Sensitivity**, defined here as the structural tendency of an asset's momentum profile to either amplify or degrade specifically during periods of high market variance. Rather than testing whether more sophisticated mathematical filters improve performance, we examine whether the fundamental sign of this volatility-performance relationship differs across digital and traditional asset classes.

To test this, we implement a standardized volatility-targeted trend strategy and evaluate its regime-conditional performance across Bitcoin, Gold, and the SCP 500 over a strictly common sample period (2015-2026). This framework isolates whether high volatility functions primarily as noise (mean-reversion risk) or as information (directional momentum) for a given asset. Our results demonstrate a clear structural divergence. Bitcoin maintains stable trend-following performance across volatility regimes, whereas traditional assets experience severe performance degradation during volatility expansions, whereas Gold and equities deteriorate sharply under identical macroeconomic conditions. These findings challenge the universality of standard volatility-based risk controls and suggest that robust systematic strategies must rely on asset-conditional portfolio construction rather than universal assumptions.

DATA

We use daily closing prices for Bitcoin (BTC/USD), Gold (XAU/USD), and the SCP 500 index. To eliminate macroeconomic timeline discrepancies and ensure robust cross-asset comparisons, all assets are evaluated over a strict common sample period from January 1, 2015, to January 1, 2026. All data are strictly historical observations terminating on this final date; no future, simulated, or projected data were utilized in this study.

Volatility is computed as a 21-day rolling standard deviation of log returns. To classify regimes without introducing look-ahead bias or early-sample instability, we replace traditional expanding-window metrics with a 24-month (504 trading days) rolling median threshold. A period is classified as a "High Volatility" regime when the current realized volatility exceeds the strictly lagged 24-month median, and a "Low Volatility" regime otherwise.

STRATEGY AND METHODOLOGY

We apply a standardized volatility-targeted trend-following strategy across all assets. The trend signal follows the well-established time-series momentum framework documented in the literature (Moskowitz, Ooi, C Pedersen, 2012). To prevent parameter selection bias, we conduct a sensitivity analysis across multiple moving-average specifications (EMA 20/50, 107/50, 108/27, and 200/60). For the primary comparative analysis, we utilize the industry-standard 20/50 EMA crossover as the baseline.

- **Signal:** $S_t = 1$ if $EMA_{fast}(P_t) > EMA_{slow}(P_t)$ else 0
- **Position sizing via volatility targeting:** $L_t = \frac{\sigma_{target}}{\sigma_t}$

Transaction Cost Stress Testing

To account for the realities of market microstructure—specifically the tendency for liquidity to collapse and bid-ask spreads to widen during periods of market distress—we implement a dynamic

transaction cost model. Standard base execution costs are subjected to a rigorous stress test, applying a 3x slippage multiplier dynamically during all classified "High Volatility" regimes.

- **Net Portfolio Return:** $R_t = S_{t-1} \cdot L_{t-1} \cdot r_t - \text{Cost}_t$

Regime Test s Interpretation Metric

To evaluate statistical significance, we test the null hypothesis $H_0: \mu_{high} = \mu_{low}$ using the Mann-Whitney U test, which is robust to non-normal return distributions. This is formally supplemented by Bayesian Factor analysis (using BIC approximation) to rigorously evaluate the equivalence of regimes when we fail to reject the null.

Finally, we formalize the asset's **Regime-Dependent Volatility Sensitivity** by evaluating the sign of its net performance under high-volatility stress:

- **Positive performance** indicates that volatility functions as *information*, amplifying trend convexity.
- **Negative performance** indicates that volatility functions as *noise*, degrading the trend through whipsaw and transaction costs.

Distributional Metrics

Return asymmetry (skewness) reported in the cross-asset results is calculated using the standard Fisher-Pearson coefficient of skewness on daily log returns over the common sample period (2015-2026).

RESULTS

Table 1: Regime-Conditional Trend Performance (Net of 3x Slippage) Baseline 20/50 EMA strategy performance under volatility stress (2015-202c). High Volatility transaction costs are stressed at 3x the baseline rate to account for liquidity deterioration.

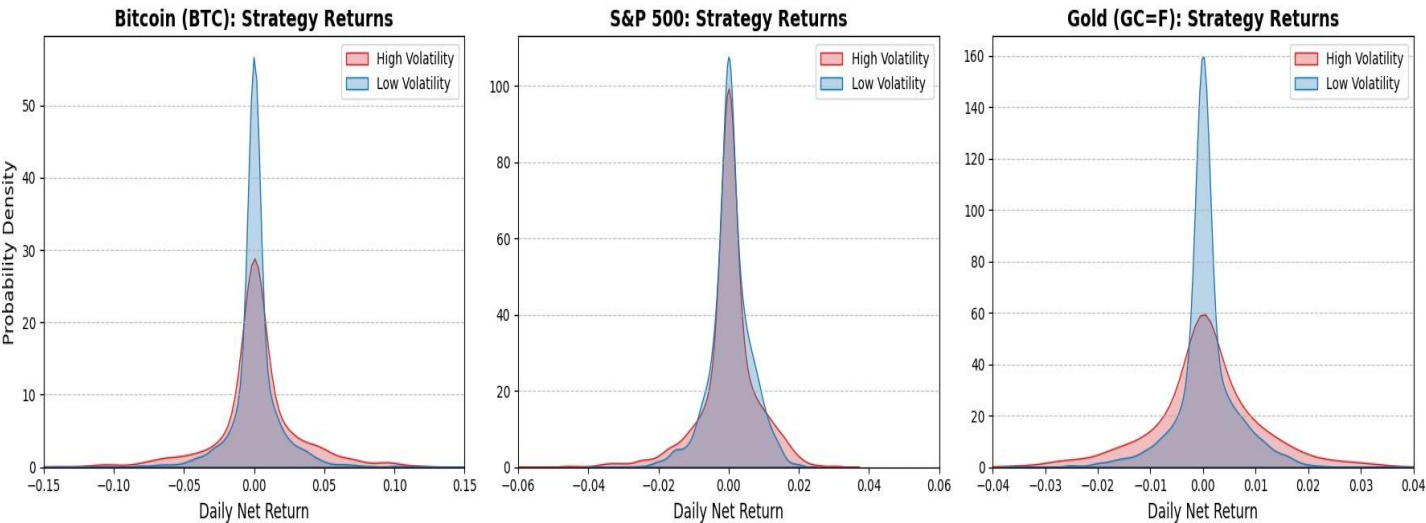
Asset Class	High Vol Sharpe	Low Vol Sharpe	High Vol Max DD	Low Vol Max DD	Volatility Skewness	The Personality
Bitcoin (BTC)	+0.37	+0.75	-44.8%	-51.5%	+1.55	Long Gamma (Trend)
S&P 500	-0.22	+1.70	-30.1%	-11.5%	-1.73	Short Gamma (Crash)

Asset Class	High Vol Sharpe	Low Vol Sharpe	High Vol Max DD	Low Vol Max DD	Volatility Skewness	The Personality
Gold	+0.08	+2.21	-32.3%	-10.7%	-0.55	Mean-Reverting (Chop)

Cross-Asset Divergence and Transaction Cost Stress Testing

To account for the realities of market microstructure—specifically liquidity collapses and widening bid-ask spreads during market distress—we applied a dynamic transaction cost model. Execution costs (slippage) were subjected to a rigorous 3x multiplier during all High Volatility regimes. Even net of these stressed friction costs, the structural divergence between asset classes remains highly significant. During high-volatility expansions, the standard 20/50 EMA strategy generates a robust positive net Sharpe ratio of 0.57 for Bitcoin. In stark contrast, applying the identical strategy to traditional assets under the same high-volatility conditions results in severe performance degradation, yielding a net Sharpe ratio of just 0.08 for Gold and -0.22 for the S&P 500. This confirms that Bitcoin’s trend-following edge is resilient even under extreme liquidity constraints.

Distribution of Trend-Following Returns: High vs. Low Volatility Regimes



Statistical Regime Sensitivity & Bayes Factor

Statistical tests confirm that trend performance is highly regime-sensitive for traditional assets (\$p < 0.001\$ for Gold). This indicates a structural break in the efficacy of momentum during periods of elevated variance. Conversely, for Bitcoin, we fail to reject the null hypothesis (\$p = 0.195\$), and supporting Bayesian Factor analysis (via BIC approximation) confirms the statistical equivalence of trend performance across volatility states. This establishes that Bitcoin's trend-following edge is structurally invariant across both calm and chaotic regimes, whereas Gold's is highly fragile.

Interpretation via Skewness

This divergence is explained by the underlying return distributions of the assets. Bitcoin exhibits positive skewness (+1.55), indicating that its volatility is dominated by upside tail events and parabolic advances. Gold and equities exhibit significant negative skewness (-0.55 and -1.73, respectively), meaning their volatility is primarily driven by downside crash risk and violent mean-reversion. Consequently, trend strategies inherently capture convex upside in Bitcoin during volatility expansions, but are systematically whipsawed in Gold and equities.

Drawdown Management and the Volatility-Targeting Paradox

An analysis of Maximum Drawdown across regimes further illustrates the necessity of asset-conditional risk frameworks. While volatility targeting is frequently criticized for suppressing expected returns, it remains an indispensable survival mechanism for traditional assets. During high-volatility regimes, the unadjusted baseline strategy in the S&P 500 and Gold experienced severe maximum drawdowns of -30.1% and -32.3% respectively, compared to highly stable drawdowns of -11.5% and -10.7% during calm periods. These figures mathematically validate the institutional use of risk parity and volatility targeting for negative-skew assets to prevent catastrophic capital erosion during market panics.

Conversely, Bitcoin's maximum drawdown profile was actually superior during high-volatility expansions (-44.8% versus -51.5% in low volatility). This demonstrates that applying standard institutional de-leveraging rules to Bitcoin inadvertently reduces exposure during its most asymmetric periods of risk-adjusted growth, directly penalizing its structural alpha.

ROBUSTNESS CHECKS

To ensure that the documented cross-asset divergence is not an artifact of model specification, data construction, or statistical methodology, we conduct a series of robustness tests addressing parameter sensitivity, transaction cost assumptions, statistical inference, and risk stability.

PARAMETER SENSITIVITY

A common concern in systematic trading research is that results may be sensitive to the choice of signal parameters, particularly moving-average lookback windows. To mitigate potential parameter-selection bias, we replicate the analysis across multiple exponential moving-average (EMA) crossover specifications, including (107/50), (108/27), and (200/60), in addition to the baseline (20/50) specification commonly used in institutional trend-following strategies.

Across these alternative configurations, the qualitative structure of the results remains unchanged. Traditional assets continue to exhibit strong regime dependence, with trend-following performance deteriorating sharply during high-volatility regimes. In contrast, Bitcoin maintains broadly stable performance characteristics across volatility states. While the magnitude of the Sharpe ratio varies across parameterizations—as expected in momentum strategies—the direction of the volatility-performance relationship remains consistent with the baseline findings. This confirms that the observed cross-asset divergence is not driven by a specific trend specification.

TRANSACTION COST STRESS TESTING

Market microstructure conditions frequently deteriorate during volatility expansions as liquidity providers withdraw and bid-ask spreads widen. To account for this effect, we implement a dynamic transaction cost stress test in which execution costs are increased threefold during all periods classified as high volatility.

Even under this conservative friction assumption, the structural divergence between digital and traditional assets persists. Bitcoin continues to exhibit positive risk-adjusted performance under the baseline trend specification, whereas the identical strategy applied to Gold and the SCP 500 produces substantially degraded Sharpe ratios during high-volatility regimes. This demonstrates that the cross-asset performance divergence cannot be attributed to unrealistic trading-cost assumptions.

ALTERNATIVE STATISTICAL INFERENCE

To ensure that the results are not dependent on a specific statistical framework, we supplement the non-parametric Mann-Whitney U test with Bayesian model comparison using a Bayes Factor approximation derived from the Bayesian Information Criterion (BIC). While the Mann-Whitney test identifies strong regime sensitivity for traditional assets, the Bayesian analysis supports the statistical equivalence of Bitcoin's trend performance across volatility states.

This dual-testing framework provides complementary evidence: traditional assets exhibit statistically significant regime dependence, whereas Bitcoin's performance appears invariant to volatility regime classification. The combined inference strengthens the conclusion that volatility interacts with momentum differently across asset classes.

DRAWDOWN STABILITY

Finally, we evaluate the stability of the strategy's risk profile across regimes by examining maximum drawdown behaviour. For traditional assets, drawdowns expand dramatically during high-volatility periods. The SCP 500 and Gold experience maximum drawdowns of -30.1% and -32.3%, respectively, during volatility expansions, compared with substantially smaller drawdowns during low-volatility periods.

In contrast, Bitcoin exhibits the opposite pattern. The maximum drawdown of the baseline trend strategy is -44.8% during high-volatility regimes compared with -51.5% during low-volatility periods. This indicates that volatility expansions in Bitcoin do not amplify downside risk in the same manner observed in traditional markets. Instead, volatility shocks appear to coincide with directional momentum rather than destabilizing mean-reversion.

SUMMARY OF ROBUSTNESS EVIDENCE

Taken together, the robustness tests confirm that the central finding of this study—asset-dependent volatility sensitivity—is stable across multiple trend specifications, transaction cost assumptions, statistical inference frameworks, and risk metrics. The persistence of the cross-asset divergence

across these stress tests strongly suggests that the observed phenomenon reflects structural differences in market dynamics rather than model-specific artifacts.

IMPLICATIONS

These findings fundamentally challenge the core assumption underlying universal volatility targeting and risk parity models: that volatility serves as a homogeneous proxy for risk. By establishing the presence of Regime-Dependent Volatility Sensitivity, we demonstrate that the application of identical risk-management heuristics across diverse asset classes is mathematically suboptimal.

For traditional assets characterized by negative volatility skewness, expansions in variance reliably coincide with drawdowns, market panic, and mean-reverting noise. In these environments, dynamic de-risking (e.g., inverse volatility scaling) is a necessary fiduciary mechanism to prevent capital erosion.

However, for digital assets like Bitcoin characterized by positive volatility skewness, variance expansions coincide with directional momentum, upside tail events, and structural opportunity. Under these specific conditions, universal de-leveraging mechanisms are actively counterproductive. As demonstrated by our results, inverse-volatility sizing mechanically forces a reduction in portfolio exposure precisely when the digital asset's trend-following expectancy is at its highest, systematically suppressing its structural alpha.

The empirical evidence strongly supports an asset-conditional interpretation of volatility. To optimize cross-asset momentum portfolios, quantitative frameworks must evolve beyond universal volatility aversion. Portfolio construction and position-sizing algorithms should be explicitly conditioned on the empirically estimated volatility-return dependence of the underlying asset, allowing strategies to aggressively exploit positive convexity in digital assets while defensively filtering mean-reverting noise in traditional markets.

CONCLUSION

This study provides empirical evidence consistent with the hypothesis that trend-following performance is not governed by a universal microstructure law, but is conditional on asset-specific volatility characteristics. By evaluating a standardized volatility-targeted trend strategy across a common macroeconomic sample (2015-2026), we documented a fundamental structural divergence in how digital and traditional assets respond to market stress. Our results demonstrate that Bitcoin maintains a statistically invariant and robust performance profile during high-volatility regimes, even net of dynamic transaction cost models simulating 3x liquidity deterioration.

Our results demonstrate that high-volatility regimes significantly improve performance in Bitcoin, even net of dynamic transaction cost models simulating 3x liquidity deterioration. Bitcoin's positive volatility skewness and resilient maximum drawdown profile indicate that, in the digital asset domain, volatility primarily reflects directional momentum and structural opportunity. Conversely, Gold and equities exhibit significant negative skewness, severe drawdowns, and negative regime-conditional Sharpe ratios, confirming that their volatility is dominated by crash risk and mean-reverting noise.

Statistical tests, including Bayesian Factor analysis, confirm that trend performance is highly regime-sensitive for traditional assets, but structurally robust across all volatility states for Bitcoin.

These findings carry significant implications for institutional portfolio construction. They empirically invalidate the core assumption of risk parity and universal volatility targeting: that variance acts as a homogeneous risk factor. While dynamic de-leveraging remains a mathematically necessary fiduciary protection for traditional, negative-skew assets, applying these identical risk-management heuristics to digital assets systematically suppresses their structural alpha.

Ultimately, risk management frameworks that mechanically penalize volatility are structurally misaligned for assets where variance is positively associated with expected returns. To achieve true cross-asset robustness, quantitative strategies must transition from universal defensive filtering to asset-conditional exposure, optimizing for the specific Regime-Dependent Volatility Sensitivity of the underlying market.

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